StockEvent Lens — Batch 003 Intelligence Dashboard

Articles 011–015 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_03 (Articles 011-015)`

TEST 011
Original — 100% Base
Nvidia customer concentration rises; CFO Colette Kress introduces neocloud 'get paid twice' strategy.

TEST 012
Repeat — 96% Overlap
Re-syndicated wire report on Nvidia's neocloud approach and customer concentration with zero new facts.

TEST 013
Update — +23% New Info
Introduces monthly utilization reporting statements starting in October for neocloud program.

TEST 014
Conflict — Dispute ($18B)
Contradicts commitment balance: claims neocloud commitments totaled $18 billion on July 26.

TEST 015
Different — 0% Overlap
Unrelated Nvidia desktop graphics adapter recall over cooling-fan fault triggering Group 006.

Technical Analysis: Dense Vector Blindness vs. Numerical Commitment Conflict Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *numerical fidelity*. Changing monetary figures (e.g., $36 billion vs. $18 billion commitment balance in test_014) yields high vector similarity (~0.91) because both texts share identical surrounding context tokens ("neocloud model", "July 26", "commitments", "Nvidia").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard numerical or monetary balance dispute.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "$36 billion" and "$18 billion" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this commitment balance conflict.

2. LLM Evaluation Override
To prevent missing critical financial statement disputes, the LLM reasoning layer performs explicit attribute-value matching, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

Article Relationship & Percentage Breakdown (Batch 003)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_011_original`	Original (%5=1)	Nvidia customer concentration, Colette Kress, neocloud model, $36B commitments	100%	Baseline	Establishes breaking news frame on Nvidia's neocloud strategy and $36B commitment benchmark.
`test_012_repeat`	Repeat (%5=2)	Nvidia sales concentration, neocloud approach, $36 billion commitments on July 26	96%	0% New Facts	Restates customer concentration and neocloud details with zero new factual developments.
`test_013_update`	Update (%5=3)	**Monthly utilization statements**, September usage, October issuance	85%	+22% Operational Facts	Advances event state by introducing monthly utilization reporting statements starting in October.
`test_014_conflict`	Conflict (%5=4)	**$18 billion commitment dispute**, July 26 balance contest, competing account	91% Match	$\Delta F_{contradiction}$ = 1.0	Directly disputes the $36 billion commitment figure, asserting commitments were $18 billion on July 26.
`test_015_different`	Different (%5=0)	Nvidia graphics adapter recall, cooling-fan manufacturing defect, serial number	0%	Group 006 Trigger	Zero domain overlap; triggers standalone Group 006 hardware manufacturing issue cluster.

Group 005 — Consolidated Event Summary

4 Articles (011–014)

NVIDIA CFO Colette Kress outlined a 'neocloud' strategy allowing the company to 'get paid twice'—earning on GPU hardware sales and sharing compute rental revenues—to mitigate customer concentration where three customers generate 44% ($78.2B) of revenue (Articles 011–012). To track this, NVIDIA introduced **monthly utilization statements launching in October to separate third-party compute usage** (Article 013). However, reporting sources dispute the historical commitment balance as of July 26: initial reports cited $36 billion, whereas a competing report claims total outstanding commitments were **$18 billion** (**Article 014** ; $\Delta F_{contradiction} = 1.0$).

✓ Update Preserved: Monthly Utilization Reporting
✓ Conflict Preserved: $36B vs $18B Commitment Balance
✓ Duplicate Removed: Article 012 Consolidated

Group 006 — Standalone Summary

1 Article (015)

NVIDIA has announced a product recall for one batch of desktop graphics adapters due to a cooling-fan manufacturing defect. Affected units will be identified by serial number, and a replacement process will begin next week (Article 015).

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 005

Batch 003 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

`test_011_original_group_005_3.json` — Group 005

```
{
  "id": 61770964,
  "headline": "Nvidia's Customer Concentration Problem
  "symbols": ["NVDA"],
  "summary": "Nvidia’s customer concentration is risin
}
```

Establishes neocloud strategy & $36B baseline.

`test_013_update_group_005_3.json` — Group 005

```
{
  "id": 990000013,
  "headline": "Nvidia Begins Monthly Utilization Repor
  "symbols": ["NVDA"],
  "summary": "Nvidia is adding monthly utilization sta
}
```

Adds monthly utilization statements in October.

`test_014_conflict_group_005_3.json` — Group 005

```
{
  "id": 990000014,
  "headline": "Competing Account Puts Nvidia Neocloud (
  "symbols": ["NVDA"],
  "summary": "A competing report says commitments unde
}
```

Disputes $36B vs $18B commitment balance.